

JEE MAIN REVISION MASTER: P-BLOCK ELEMENTS 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 25 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	P Block Elements — Applied Cases, Reactions & Traps	14 PYQs	Applied Topics
Moderate	P Block Elements — Standard Calculations & Problem Archetypes	7 PYQs	Quantitative Analysis
Moderate	P Block Elements — Core Principles & Governing Laws	5 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
P Block Elements — Applied Cases, Reactions & Traps	N_2O , NH_3 , (H_3PO_4)
P Block Elements — Standard Calculations & Problem Archetypes	$CaSO_4 \cdot \frac{1}{2}H_2O$, $BaCl_2$, $_{4}$
P Block Elements — Core Principles & Governing Laws	$NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$? , (H_3PO_2) , $AgNO_3$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 P Block Elements — Applied Cases, Reactions & Traps (High Yield — 14 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of P Block Elements.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): " NO_2 gas is..." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "Which of the following compound does not give dinitrogen on heating?" [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "The correct decreasing order of basicity of hydrides of group- 15 elements is..." [Ans: N/A]

3.2 P Block Elements — Standard Calculations & Problem Archetypes (Moderate Yield — 7 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in P Block Elements.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which of the following halides cannot be hydrolysed?..." [Ans: N/A]
- [2023 Shift] Q#4 (MCQ): "What is the oxidation number of S in $H_2S_2O_8$?..." [Ans: N/A]
- [2022 Shift] Q#4 (MCQ): "Which of the following hydrides is electron deficient?..." [Ans: N/A]

3.3 P Block Elements — Core Principles & Governing Laws (Moderate Yield — 5 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in P Block Elements.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in P Block Elements.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in P Block Elements PYQs.

Representative Source PYQs:

- [2024 Shift] Q#3 (MCQ): "A metalloid is..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "Which one of the following oxoacids of phosphorus can reduce $AgNO_3$ to metallic silver?..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "The strong reducing property of hypophosphorus acid is due to..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#2)

NO_2 gas is

- (A) colourless, neutral
- (B) colourless, acidic
- (C) brown, acidic
- (D) brown neutral

Correct Answer: None

Solution Analysis: NO_2 gas is brown and acidic, so the correct choice is: Option C : brown, acidic NO_2 stands for nitrogen dioxide, which is a reddish-brown gas known for its characteristic pungent smell. It is one of the primary pollutants resulting from the combustion of fossil fuels and is a contributor to atmospheric reactions that produce smog and acid rain. As an acidic gas, NO_2 can dissolve in water to form nitric and nitrous acids, contributing to environmental issues such as acid rain.

Benchmark Problem 2: KCET 2024 (Q#3)

A metalloid is

- (A) Bi
- (B) Sb
- (C) P
- (D) Se

Correct Answer: None

Solution Analysis: A metalloid can be defined as chemical elements whose physical and chemical properties lies in between metal and non-metal. Among the given option only antimony (Sb) is metalloid.

Benchmark Problem 3: KCET 2023 (Q#4)

What is the oxidation number of S in $\text{H}_2\text{S}_2\text{O}_8$?

- (A) +5
- (B) +4
- (C) +7
- (D) +6

Correct Answer: None

Solution Analysis: The structure of $\text{H}_2\text{S}_2\text{O}_8$ is as follows Let the oxidation state of sulphur be x . Then, $2x + 2 + 2(-1) + 6(-2) = 0 \therefore x = +6$ Thus, the oxidation state of S in $\text{H}_2\text{S}_2\text{O}_8$ is +6 .